

AI-Powered Enterprise E-Commerce Platform

Database Requirements Specification

Document Information

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Table of Contents

- 1. Introduction
- 2. Purpose
- 3. Database Objectives
- 4. Database Architecture
- 5. Database Selection
- 6. Data Requirements
- 7. Entity Requirements
- 8. Relationship Requirements
- 9. Data Integrity Requirements
- 10. Security Requirements
- 11. Performance Requirements
- 12. AI Data Requirements
- 13. Backup and Recovery Requirements
- 14. Data Retention Requirements
- 15. Future Enhancements
- 16. Conclusion

17. References

1. Introduction

The Database Requirements Specification defines the data storage requirements of the AI-Powered Enterprise E-Commerce Platform.

The database is responsible for storing:

- customer information,
- product information,
- transactions,
- AI datasets,
- analytics data,
- and operational records.

A well-designed database ensures consistency, security, scalability, and high performance.

2. Purpose

The purpose of this document is to:

- define database requirements,
 - identify entities and relationships,
 - establish integrity constraints,
 - support future database design,
 - improve scalability and maintainability.
-

3. Database Objectives

The database should:

DBO-01

Provide secure storage.

DBO-02

Support high transaction volumes.

DBO-03

Maintain data integrity.

DBO-04

Support AI analytics requirements.

DBO-05

Provide scalability.

DBO-06

Support future expansion.

4. Database Architecture

The system will use a multi-database approach.

Primary Transactional Database

Technology:

PostgreSQL

Purpose:

Store transactional data.

Cache Database

Technology:

Redis

Purpose:

Store temporary data.

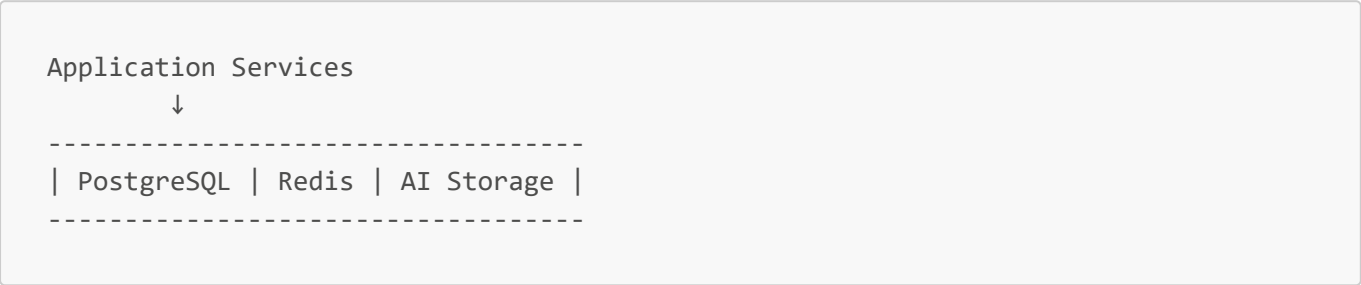
AI Data Storage

Purpose:

Store:

- datasets,
- feature data,
- model metadata,
- predictions.

Database Architecture Overview



5. Database Selection Justification

PostgreSQL

Reasons:

- ACID compliance
- Reliability
- Strong relational capabilities
- Excellent indexing support
- JSON support
- Industry adoption

Redis

Reasons:

- Extremely fast
- Ideal for caching
- Session storage
- Recommendation caching

6. Data Categories

The system manages several categories of data.

Master Data

Examples:

- Users
 - Products
 - Categories
 - Roles
-

Transactional Data

Examples:

- Orders
 - Payments
 - Inventory updates
-

Analytical Data

Examples:

- Customer behavior
 - Sales reports
 - Product performance
-

AI Data

Examples:

- Recommendation datasets
 - Fraud datasets
 - Forecasting datasets
-

System Data

Examples:

- Logs
 - Notifications
 - Audit records
-

7. Entity Requirements

The following major entities have been identified.

7.1 User Entity

Purpose:

Store customer and administrator information.

Attributes:

- user_id
- first_name
- last_name
- email
- password_hash
- phone
- status
- created_at
- updated_at

Primary Key:

user_id

7.2 Role Entity

Purpose:

Store RBAC information.

Attributes:

- role_id
 - role_name
 - description
-

7.3 Address Entity

Purpose:

Store delivery addresses.

Attributes:

- address_id
- user_id
- address_line_1
- city
- postal_code

- country
-

7.4 Product Entity

Purpose:

Store product information.

Attributes:

- product_id
 - product_name
 - description
 - price
 - stock_quantity
 - category_id
 - status
-

7.5 Category Entity

Attributes:

- category_id
 - category_name
 - description
-

7.6 Product Image Entity

Attributes:

- image_id
 - product_id
 - image_url
-

7.7 Cart Entity

Attributes:

- cart_id
 - user_id
 - created_at
-

7.8 Cart Item Entity

Attributes:

- cart_item_id
 - cart_id
 - product_id
 - quantity
-

7.9 Wishlist Entity

Attributes:

- wishlist_id
 - user_id
-

7.10 Order Entity

Attributes:

- order_id
 - user_id
 - order_date
 - total_amount
 - order_status
-

7.11 Order Item Entity

Attributes:

- order_item_id
 - order_id
 - product_id
 - quantity
 - unit_price
-

7.12 Payment Entity

Attributes:

- payment_id

- order_id
 - payment_method
 - payment_status
 - transaction_reference
-

7.13 Inventory Entity

Attributes:

- inventory_id
 - product_id
 - available_quantity
 - reserved_quantity
-

7.14 Review Entity

Attributes:

- review_id
 - user_id
 - product_id
 - rating
 - comment
-

7.15 Notification Entity

Attributes:

- notification_id
 - user_id
 - message
 - notification_type
-

7.16 Audit Log Entity

Purpose:

Track system activities.

Attributes:

- log_id
- actor_id

- action
- timestamp
- ip_address

7.17 Fraud Alert Entity

Attributes:

- fraud_id
- order_id
- risk_score
- alert_status

7.18 Recommendation History Entity

Attributes:

- recommendation_id
- user_id
- recommended_products
- generated_at

7.19 Sales Forecast Entity

Attributes:

- forecast_id
- product_id
- predicted_quantity
- forecast_date

8. Relationship Requirements

User → Address

Relationship:

One-to-Many

User → Orders

Relationship:

One-to-Many

Product → Category

Relationship:

Many-to-One

Order → Order Items

Relationship:

One-to-Many

Product → Reviews

Relationship:

One-to-Many

Product → Inventory

Relationship:

One-to-One

User → Cart

Relationship:

One-to-One

User → Wishlist

Relationship:

One-to-One

User → Notifications

Relationship:

One-to-Many

9. Conceptual ER Model

```
User
↓
Order
↓
OrderItem
↓
Product
↓
Category
```

Additional relationships:

```
User → Review → Product
User → Cart → CartItem
User → Wishlist
Product → Inventory
Order → Payment
Order → FraudAlert
```

10. Normalization Requirements

The database should satisfy:

First Normal Form (1NF)

No repeating groups.

Second Normal Form (2NF)

No partial dependencies.

Third Normal Form (3NF)

No transitive dependencies.

BCNF should be considered where applicable.

11. Data Integrity Requirements

Entity Integrity

Every table shall contain a primary key.

Referential Integrity

Foreign keys shall maintain relationships.

Domain Integrity

Examples:

- Ratings between 1–5.
 - Prices cannot be negative.
 - Stock quantities cannot be negative.
-

Business Integrity Rules

Examples:

- Orders cannot exist without users.
 - Payments cannot exist without orders.
-

12. Database Constraints

Examples:

Unique Constraints

- Email addresses
 - Transaction references
-

Check Constraints

- `rating >= 1`
 - `rating <= 5`
 - `price >= 0`
-

Foreign Key Constraints

Maintain referential consistency.

13. Indexing Requirements

Indexes should be created for:

- email
 - product_name
 - category_id
 - order_date
 - payment_status
 - created_at
-

Full Text Search Indexes

For:

- product search
 - recommendation queries
-

14. Performance Requirements

Response Time

Most queries:

< 2 seconds.

Concurrent Transactions

Support:

1,000+ users.

Scalability

Support millions of records.

15. Security Requirements

Authentication Data Security

Passwords must be:

- hashed
 - salted
-

Encryption Requirements

Sensitive information should be encrypted.

Examples:

- personal information
 - payment references
-

Access Control

Role-based access to data.

Audit Logging

All administrative activities shall be recorded.

16. Backup Requirements

Backup Frequency

Daily backups.

Incremental Backups

Every few hours.

Backup Storage

Separate secure locations.

17. Recovery Requirements

Recovery Time Objective (RTO)

Less than 4 hours.

Recovery Point Objective (RPO)

Less than 30 minutes.

18. Data Retention Requirements

Examples:

Audit Logs

Retain for:

2 years.

Order History

Retain for:

5 years.

Recommendation History

Retain according to business needs.

19. AI Data Requirements

AI services require additional data.

Recommendation Data

Required:

- clicks
 - purchases
 - product views
 - ratings
-

Fraud Detection Data

Required:

- transaction patterns
- payment information

- user behavior
-

Forecasting Data

Required:

- historical sales
 - seasonal information
 - inventory history
-

Customer Segmentation Data

Required:

- purchase frequency
 - customer spending
 - browsing patterns
-

20. Data Pipeline Requirements

```
graph TD; A[Transactional Data] --> B[Data Cleaning]; B --> C[Feature Engineering]; C --> D[Model Training]; D --> E[Predictions];
```

21. Database Risks

Possible risks:

- Data corruption
- Data breaches
- Performance bottlenecks
- Inconsistent datasets
- AI data quality issues

Mitigation:

- backups,

- monitoring,
 - indexing,
 - validation,
 - security controls.
-

22. Future Enhancements

Possible future improvements:

- Database sharding
 - Read replicas
 - Data warehouse
 - Data lake
 - Event sourcing
 - Real-time analytics pipelines
-

23. Conclusion

The proposed database architecture provides:

- reliability,
- scalability,
- security,
- maintainability,
- and strong support for AI services.

The database design establishes a solid foundation for building an enterprise-grade intelligent e-commerce platform.

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